KHOULOUD BOUAZIZ

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linkedin

PROFILE

Data and Decision Systems Engineer, specialized in Artificial Intelligence, LLMs, and Intelligent Automation. Passionate about transforming data into smart, impactful solutions by combining technical expertise, analytical thinking, and innovation within agile environments.

EDUCATION

National School of Electronics and Telecommunications of Sfax (ENET'Com) **2022 - 2025**

Engineering Degree in Data Engineering & Decision Systems

Faculty of Sciences of Sfax (FSS) **2020 - 2022**

Preparatory Classes – Physics & Chemistry Track

PROFESSIONAL EXPERIENCE

Final study project : Smart QA – Intelligent Test Automation Platform **Feb 2025 - June2025**

Freedom Of Dev – Sfax, Tunisia | Intern

- Designed and developed a full-stack test automation platform using React, FastAPI, and Quarkus.
- Integrated Playwright for real-time test execution and leveraged an LLM to automatically generate test cases and test scripts from Azure DevOps user stories, enhancing test coverage and efficiency.
- Implemented Microsoft Entra ID for secure authentication and role-based access control.
- Built dynamic dashboards and reports for KPI tracking, test traceability, and decision support.

Real-Time Anomaly Detection in Water Flow Systems **July 2024 - August 2024**

Digital Research Center of Sfax, Tunisia | Intern

- Built a real-time anomaly detection system using Isolation Forest, LSTM, and LOF for water flow monitoring.
- Improved detection accuracy and enabled early identification of leaks and abnormal pressure events.

Optimization Algorithm for Solar Panel Cleaning Robots **June 2023 - July 2023**

Enova Robotics – Sousse, Tunisia | Intern

- Developed a Python algorithm to optimize solar panel cleaning robot trajectories, enhancing efficiency and enabling obstacle avoidance.

ACADEMIC PROJECTS

Evaluation and Mitigation of Ethical Bias in Machine Learning Models

Final Year Project at ENET'Com

- Conducted ethical bias assessment and implemented mitigation strategies for machine learning models.
- Utilized AIF360 and scikit-learn for model development and evaluation, applied LIME and SHAP for interpretability, and deployed the project with Streamlit.

Emotion Classification with NLP

- Development and deployment of an emotion classification model using NLTK for text preprocessing, TF-IDF and LSTM for deep learning-based classification, and Streamlit for deployment.

Automated testing with Robot Framework

- Automation of functional test scenarios for the Amazon website (search, add to cart, verification)
- using Robot Framework and Selenium.

TECHNICAL SKILLS

- **Programming: Python, Java, C, R, Scala**
- **Testing & Automation: Playwright, Selenium, Robot Framework**
- **Artificial Intelligence: Machine Learning, Deep Learning, Computer Vision, NLP,**
- **Generative AI (LLM), MLOps**
- **AI Frameworks: TensorFlow, Keras, scikit-learn, OpenCV, MLflow**
- **Back-end: Quarkus, Spring Boot, FastAPI**
- **Front-end: React, HTML, CSS, JavaScript**
- **DevOps: Azure DevOps, Git**
- **Databases: MySQL, PostgreSQL, MongoDB, NoSQL**
- **Visualization & Reporting: Power BI, Excel**
- **Operating Systems: Windows, Linux**

CERTIFICATIONS

- Microsoft Certified: Azure Data Fundamentals
- Microsoft Certified: Azure AI Fundamentals
- Certificate of Competency – Computer Vision for Industrial Inspection, NVIDIA

LANGUAGES

- Arabic-Native
- French - Proficent
- English -Fluent